Litter: regression-imputation application

Simultaneous statistical inferences of treatment-control mean differences

Dose comparison	Point estimate	95% simultaneous confidence interval	Interval length
Proposed method with shrinkage			
5 vs. 0	-3.485	(-8.122, 1.152)	9.274
50 vs. 0	-4.013	(-8.915, 0.889)	9.805
500 vs. 0	-4.744	(-9.245, -0.242)	9.003
Proposed method without shrinkage			
5 vs. 0	-3.485	(-8.018, 1.049)	9.067
50 vs. 0	-4.013	(-8.726, 0.700)	9.426
500 vs. 0	-4.744	(-9.148, -0.339)	8.809
Dunnett-type procedure with naive case deletion			
5 vs. 0	-3.123	(-6.957, 0.710)	7.667
50 vs. 0	-3.461	(-7.119, 0.197)	7.316
500 vs. 0	-4.907	(-8.964, -0.851)	8.113

Primary analysis: 40 regression imputations shared by both proposed procedures. Target expected MAR missingness is 30%; realized missingness is 26/74 = 35.14%. The covariate is standardized litter size.

Only regression was run. The uploaded script also listed PMM/PS; those comparisons were disabled to honour the user's regression-only instruction. The original seeds and REG calculations are retained.

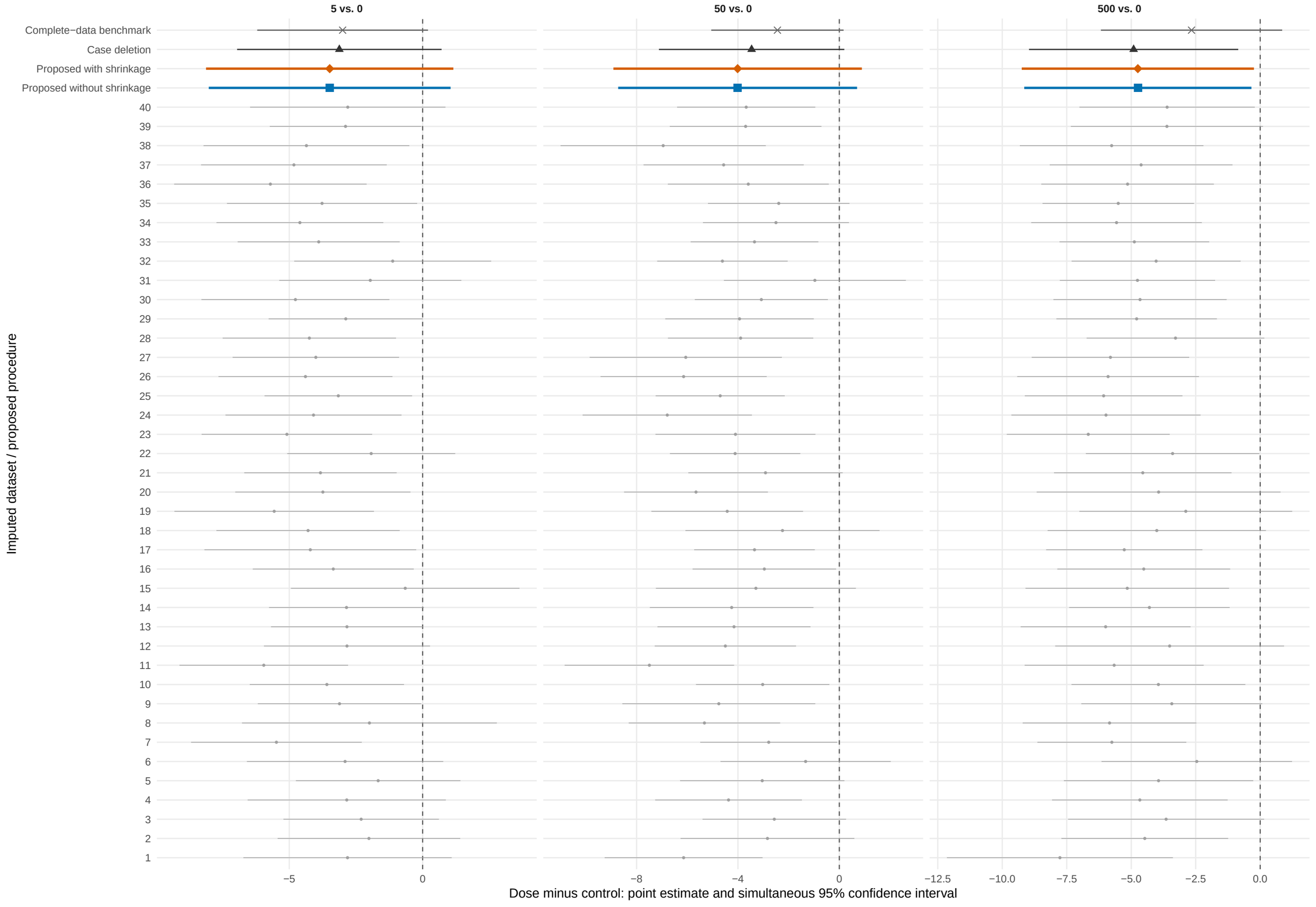
Validation check specified by the script: 500 new MAR repetitions requested. Benchmark recovery is against the fixed complete-sample treatment-control contrasts, not population-parameter coverage.

Non-shrinkage: 500 successful repetitions; joint benchmark recovery 100.00%; mean interval length 8.255.

Shrinkage: 500 successful repetitions; joint benchmark recovery 100.00%; mean interval length 8.520.

Litter: regression imputation − proposed procedures

40 regression imputations; 30% target MAR (35.14% realized missingness)



Result ● Individual imputation ■ Proposed without shrinkage ◆ Proposed with shrinkage ▲ Case deletion × Complete−data benchmark

Highlighted rows combine all 40 regression imputations using Rubin rules. Blue square: without shrinkage; orange diamond: with shrinkage.
Grey numbered rows are individual imputations. Colour identifies procedures, not significance or superiority.